附錄三　三角函數表

表 1. 三角函數表（正弦餘弦）

$$y = \sin\theta$$

θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	0.0000	0.0017	0.0035	0.0052	0.0070	0.0087	0.0105	0.0122	0.0140	0.0157
1	0.0175	0.0192	0.0209	0.0227	0.0244	0.0262	0.0279	0.0297	0.0314	0.0332
2	0.0349	0.0366	0.0384	0.0401	0.0419	0.0436	0.0454	0.0471	0.0488	0.0506
3	0.0523	0.0541	0.0558	0.0576	0.0593	0.0610	0.0628	0.0645	0.0663	0.0680
4	0.0698	0.0715	0.0732	0.0750	0.0767	0.0785	0.0802	0.0819	0.0837	0.0854
5	0.0872	0.0889	0.0906	0.0924	0.0941	0.0958	0.0976	0.0993	0.1011	0.1028
6	0.1045	0.1063	0.1080	0.1097	0.1115	0.1132	0.1149	0.1167	0.1184	0.1201
7	0.1219	0.1236	0.1253	0.1271	0.1288	0.1305	0.1323	0.1340	0.1357	0.1374
8	0.1392	0.1409	0.1426	0.1444	0.1461	0.1478	0.1495	0.1513	0.1530	0.1547
9	0.1564	0.1582	0.1599	0.1616	0.1633	0.1650	0.1668	0.1685	0.1702	0.1719
10	0.1736	0.1754	0.1771	0.1788	0.1805	0.1822	0.1839	0.1857	0.1874	0.1891
11	0.1908	0.1925	0.1942	0.1959	0.1977	0.1994	0.2011	0.2028	0.2045	0.2062
12	0.2079	0.2096	0.2113	0.2130	0.2147	0.2164	0.2181	0.2198	0.2215	0.2232
13	0.2249	0.2266	0.2283	0.2300	0.2317	0.2334	0.2351	0.2368	0.2385	0.2402
14	0.2419	0.2436	0.2453	0.2470	0.2487	0.2504	0.2521	0.2538	0.2554	0.2571
15	0.2588	0.2605	0.2622	0.2639	0.2656	0.2672	0.2689	0.2706	0.2723	0.2740
16	0.2756	0.2773	0.2790	0.2807	0.2823	0.2840	0.2857	0.2874	0.2890	0.2907
17	0.2924	0.2940	0.2957	0.2974	0.2990	0.3007	0.3024	0.3040	0.3057	0.3074
18	0.3090	0.3107	0.3123	0.3140	0.3156	0.3173	0.3190	0.3206	0.3223	0.3239
19	0.3256	0.3272	0.3289	0.3305	0.3322	0.3338	0.3354	0.3371	0.3387	0.3404
20	0.3420	0.3437	0.3453	0.3469	0.3486	0.3502	0.3518	0.3535	0.3551	0.3567
21	0.3584	0.3600	0.3616	0.3632	0.3649	0.3665	0.3681	0.3697	0.3714	0.3730
22	0.3746	0.3762	0.3778	0.3795	0.3811	0.3827	0.3843	0.3859	0.3875	0.3891
23	0.3907	0.3923	0.3939	0.3955	0.3971	0.3987	0.4003	0.4019	0.4035	0.4051
24	0.4067	0.4083	0.4099	0.4115	0.4131	0.4147	0.4163	0.4179	0.4194	0.4210
25	0.4226	0.4242	0.4258	0.4274	0.4289	0.4305	0.4321	0.4337	0.4352	0.4368
26	0.4384	0.4399	0.4415	0.4431	0.4446	0.4462	0.4478	0.4493	0.4509	0.4524
27	0.4540	0.4555	0.4571	0.4586	0.4602	0.4617	0.4633	0.4648	0.4664	0.4679
28	0.4695	0.4710	0.4725	0.4741	0.4756	0.4772	0.4787	0.4802	0.4817	0.4833
29	0.4848	0.4863	0.4879	0.4894	0.4909	0.4924	0.4939	0.4955	0.4970	0.4985
30	0.5000	0.5015	0.5030	0.5045	0.5060	0.5075	0.5090	0.5105	0.5120	0.5135
31	0.5150	0.5165	0.5180	0.5195	0.5210	0.5225	0.5240	0.5255	0.5269	0.5284
32	0.5299	0.5314	0.5329	0.5343	0.5358	0.5373	0.5388	0.5402	0.5417	0.5432
33	0.5446	0.5461	0.5476	0.5490	0.5505	0.5519	0.5534	0.5548	0.5563	0.5577
34	0.5592	0.5606	0.5621	0.5635	0.5650	0.5664	0.5678	0.5693	0.5707	0.5721
35	0.5736	0.5750	0.5764	0.5778	0.5793	0.5807	0.5821	0.5835	0.5849	0.5864
36	0.5878	0.5892	0.5906	0.5920	0.5934	0.5948	0.5962	0.5976	0.5990	0.6004
37	0.6018	0.6032	0.6046	0.6060	0.6074	0.6088	0.6101	0.6115	0.6129	0.6143
38	0.6157	0.6170	0.6184	0.6198	0.6211	0.6225	0.6239	0.6252	0.6266	0.6280
39	0.6293	0.6307	0.6320	0.6334	0.6347	0.6361	0.6374	0.6388	0.6401	0.6414
40	0.6428	0.6441	0.6454	0.6468	0.6481	0.6494	0.6508	0.6521	0.6534	0.6547
41	0.6560	0.6574	0.6587	0.6600	0.6613	0.6626	0.6639	0.6652	0.6665	0.6678
42	0.6691	0.6704	0.6717	0.6730	0.6743	0.6756	0.6769	0.6782	0.6794	0.6807
43	0.6820	0.6833	0.6845	0.6858	0.6871	0.6883	0.6896	0.6909	0.6921	0.6934
44	0.6946	0.6959	0.6972	0.6984	0.6997	0.7009	0.7021	0.7034	0.7046	0.7059
θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

餘弦$\cos\theta = \sin(90° - \theta)$

表 1. （續）三角函數表（正弦餘弦）

$$y = \sin\theta$$

θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
45	0.7071	0.7083	0.7096	0.7108	0.7120	0.7132	0.7145	0.7157	0.7169	0.7181
46	0.7193	0.7205	0.7218	0.7230	0.7242	0.7254	0.7266	0.7278	0.7290	0.7302
47	0.7313	0.7325	0.7337	0.7349	0.7361	0.7373	0.7384	0.7396	0.7408	0.7420
48	0.7431	0.7443	0.7455	0.7466	0.7478	0.7489	0.7501	0.7513	0.7524	0.7536
49	0.7547	0.7558	0.7570	0.7581	0.7593	0.7604	0.7615	0.7627	0.7638	0.7649
50	0.7660	0.7672	0.7683	0.7694	0.7705	0.7716	0.7727	0.7738	0.7749	0.7760
51	0.7771	0.7782	0.7793	0.7804	0.7815	0.7826	0.7837	0.7848	0.7858	0.7869
52	0.7880	0.7891	0.7901	0.7912	0.7923	0.7933	0.7944	0.7955	0.7965	0.7976
53	0.7986	0.7997	0.8007	0.8018	0.8028	0.8038	0.8049	0.8059	0.8070	0.8080
54	0.8090	0.8100	0.8111	0.8121	0.8131	0.8141	0.8151	0.8161	0.8171	0.8181
55	0.8191	0.8201	0.8211	0.8221	0.8231	0.8241	0.8251	0.8261	0.8271	0.8281
56	0.8290	0.8300	0.8310	0.8319	0.8329	0.8339	0.8348	0.8358	0.8368	0.8377
57	0.8387	0.8396	0.8406	0.8415	0.8424	0.8434	0.8443	0.8453	0.8462	0.8471
58	0.8480	0.8490	0.8499	0.8508	0.8517	0.8526	0.8535	0.8544	0.8554	0.8563
59	0.8572	0.8581	0.8590	0.8598	0.8607	0.8616	0.8625	0.8634	0.8643	0.8651
60	0.8660	0.8669	0.8678	0.8686	0.8695	0.8703	0.8712	0.8721	0.8729	0.8738
61	0.8746	0.8755	0.8763	0.8771	0.8780	0.8788	0.8796	0.8805	0.8813	0.8821
62	0.8829	0.8838	0.8846	0.8854	0.8862	0.8870	0.8878	0.8886	0.8894	0.8902
63	0.8910	0.8918	0.8926	0.8934	0.8941	0.8949	0.8957	0.8965	0.8973	0.8980
64	0.8988	0.8995	0.9003	0.9011	0.9018	0.9026	0.9033	0.9041	0.9048	0.9056
65	0.9063	0.9070	0.9078	0.9085	0.9092	0.9100	0.9107	0.9114	0.9121	0.9128
66	0.9135	0.9142	0.9150	0.9157	0.9164	0.9171	0.9177	0.9184	0.9191	0.9198
67	0.9205	0.9212	0.9219	0.9225	0.9232	0.9239	0.9245	0.9252	0.9259	0.9265
68	0.9272	0.9278	0.9285	0.9291	0.9298	0.9304	0.9310	0.9317	0.9323	0.9329
69	0.9336	0.9342	0.9348	0.9354	0.9361	0.9367	0.9373	0.9379	0.9385	0.9391
70	0.9397	0.9403	0.9409	0.9415	0.9421	0.9426	0.9432	0.9438	0.9444	0.9449
71	0.9455	0.9461	0.9466	0.9472	0.9478	0.9483	0.9489	0.9494	0.9500	0.9505
72	0.9511	0.9516	0.9521	0.9527	0.9532	0.9537	0.9542	0.9548	0.9553	0.9558
73	0.9563	0.9568	0.9573	0.9578	0.9583	0.9588	0.9593	0.9598	0.9603	0.9608
74	0.9613	0.9617	0.9622	0.9627	0.9632	0.9636	0.9641	0.9646	0.9650	0.9655
75	0.9659	0.9664	0.9668	0.9673	0.9677	0.9681	0.9686	0.9690	0.9694	0.9699
76	0.9703	0.9707	0.9711	0.9715	0.9720	0.9724	0.9728	0.9732	0.9736	0.9740
77	0.9744	0.9748	0.9751	0.9755	0.9759	0.9763	0.9767	0.9770	0.9774	0.9778
78	0.9781	0.9785	0.9789	0.9792	0.9796	0.9799	0.9803	0.9806	0.9810	0.9813
79	0.9816	0.9820	0.9823	0.9826	0.9829	0.9833	0.9836	0.9839	0.9842	0.9845
80	0.9848	0.9851	0.9854	0.9857	0.9860	0.9863	0.9866	0.9869	0.9871	0.9874
81	0.9877	0.9880	0.9882	0.9885	0.9888	0.9890	0.9893	0.9895	0.9898	0.9900
82	0.9903	0.9905	0.9907	0.9910	0.9912	0.9914	0.9917	0.9919	0.9921	0.9923
83	0.9925	0.9928	0.9930	0.9932	0.9934	0.9936	0.9938	0.9940	0.9941	0.9943
84	0.9945	0.9947	0.9949	0.9951	0.9952	0.9954	0.9956	0.9957	0.9959	0.9960
85	0.9962	0.9963	0.9965	0.9966	0.9968	0.9969	0.9971	0.9972	0.9973	0.9974
86	0.9976	0.9977	0.9978	0.9979	0.9980	0.9981	0.9982	0.9983	0.9984	0.9985
87	0.9986	0.9987	0.9988	0.9989	0.9990	0.9990	0.9991	0.9992	0.9993	0.9993
88	0.9994	0.9994	0.9995	0.9996	0.9996	0.9997	0.9997	0.9997	0.9998	0.9998
89	0.9998	0.9999	0.9999	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000
θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

餘弦$\cos\theta = \sin(90° - \theta)$

表 2. 三角函數表（正切餘切）

$y = \tan\theta$

θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	0.0000	0.0017	0.0035	0.0052	0.0070	0.0087	0.0105	0.0122	0.0140	0.0157
1	0.0175	0.0192	0.0209	0.0227	0.0244	0.0262	0.0279	0.0297	0.0314	0.0332
2	0.0349	0.0367	0.0384	0.0402	0.0419	0.0437	0.0454	0.0472	0.0489	0.0507
3	0.0524	0.0542	0.0559	0.0577	0.0594	0.0612	0.0629	0.0647	0.0664	0.0682
4	0.0699	0.0717	0.0734	0.0752	0.0769	0.0787	0.0805	0.0822	0.0840	0.0857
5	0.0875	0.0892	0.0910	0.0928	0.0945	0.0963	0.0980	0.0998	0.1016	0.1033
6	0.1051	0.1069	0.1086	0.1104	0.1122	0.1139	0.1157	0.1175	0.1192	0.1210
7	0.1228	0.1246	0.1263	0.1281	0.1299	0.1317	0.1334	0.1352	0.1370	0.1388
8	0.1405	0.1423	0.1441	0.1459	0.1477	0.1494	0.1512	0.1530	0.1548	0.1566
9	0.1584	0.1602	0.1620	0.1638	0.1655	0.1673	0.1691	0.1709	0.1727	0.1745
10	0.1763	0.1781	0.1799	0.1817	0.1835	0.1853	0.1871	0.1889	0.1908	0.1926
11	0.1944	0.1962	0.1980	0.1998	0.2016	0.2034	0.2053	0.2071	0.2089	0.2107
12	0.2126	0.2144	0.2162	0.2180	0.2199	0.2217	0.2235	0.2254	0.2272	0.2290
13	0.2309	0.2327	0.2345	0.2364	0.2382	0.2401	0.2419	0.2438	0.2456	0.2475
14	0.2493	0.2512	0.2530	0.2549	0.2568	0.2586	0.2605	0.2623	0.2642	0.2661
15	0.2679	0.2698	0.2717	0.2736	0.2754	0.2773	0.2792	0.2811	0.2830	0.2849
16	0.2867	0.2886	0.2905	0.2924	0.2943	0.2962	0.2981	0.3000	0.3019	0.3038
17	0.3057	0.3076	0.3095	0.3115	0.3134	0.3153	0.3172	0.3191	0.3211	0.3230
18	0.3249	0.3268	0.3288	0.3307	0.3326	0.3346	0.3365	0.3385	0.3404	0.3424
19	0.3443	0.3463	0.3482	0.3502	0.3521	0.3541	0.3561	0.3580	0.3600	0.3620
20	0.3640	0.3659	0.3679	0.3699	0.3719	0.3739	0.3759	0.3779	0.3799	0.3819
21	0.3839	0.3859	0.3879	0.3899	0.3919	0.3939	0.3959	0.3979	0.4000	0.4020
22	0.4040	0.4061	0.4081	0.4101	0.4122	0.4142	0.4163	0.4183	0.4204	0.4224
23	0.4245	0.4265	0.4286	0.4307	0.4327	0.4348	0.4369	0.4390	0.4410	0.4431
24	0.4452	0.4473	0.4494	0.4515	0.4536	0.4557	0.4578	0.4599	0.4621	0.4642
25	0.4663	0.4684	0.4706	0.4727	0.4748	0.4770	0.4791	0.4813	0.4834	0.4856
26	0.4877	0.4899	0.4921	0.4942	0.4964	0.4986	0.5008	0.5029	0.5051	0.5073
27	0.5095	0.5117	0.5139	0.5161	0.5183	0.5206	0.5228	0.5250	0.5272	0.5295
28	0.5317	0.5339	0.5362	0.5384	0.5407	0.5429	0.5452	0.5475	0.5497	0.5520
29	0.5543	0.5566	0.5589	0.5612	0.5635	0.5658	0.5681	0.5704	0.5727	0.5750
30	0.5773	0.5797	0.5820	0.5843	0.5867	0.5890	0.5914	0.5937	0.5961	0.5985
31	0.6008	0.6032	0.6056	0.6080	0.6104	0.6128	0.6152	0.6176	0.6200	0.6224
32	0.6249	0.6273	0.6297	0.6322	0.6346	0.6371	0.6395	0.6420	0.6444	0.6469
33	0.6494	0.6519	0.6544	0.6569	0.6594	0.6619	0.6644	0.6669	0.6694	0.6720
34	0.6745	0.6770	0.6796	0.6821	0.6847	0.6873	0.6898	0.6924	0.6950	0.6976
35	0.7002	0.7028	0.7054	0.7080	0.7106	0.7133	0.7159	0.7186	0.7212	0.7239
36	0.7265	0.7292	0.7319	0.7346	0.7372	0.7399	0.7426	0.7454	0.7481	0.7508
37	0.7535	0.7563	0.7590	0.7618	0.7645	0.7673	0.7701	0.7729	0.7757	0.7785
38	0.7813	0.7841	0.7869	0.7897	0.7926	0.7954	0.7983	0.8011	0.8040	0.8069
39	0.8098	0.8127	0.8156	0.8185	0.8214	0.8243	0.8273	0.8302	0.8331	0.8361
40	0.8391	0.8421	0.8450	0.8480	0.8510	0.8541	0.8571	0.8601	0.8632	0.8662
41	0.8693	0.8723	0.8754	0.8785	0.8816	0.8847	0.8878	0.8909	0.8941	0.8972
42	0.9004	0.9035	0.9067	0.9099	0.9131	0.9163	0.9195	0.9228	0.9260	0.9292
43	0.9325	0.9358	0.9390	0.9423	0.9456	0.9489	0.9523	0.9556	0.9589	0.9623
44	0.9657	0.9690	0.9724	0.9758	0.9792	0.9827	0.9861	0.9896	0.9930	0.9965
θ (degrees)	90	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

餘切 $\cot\theta = \tan(90° - \theta)$

表 2. （續）三角函數表（正切餘切）

$y = \tan\theta$

θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
45	1.0000	1.0035	1.0070	1.0105	1.0140	1.0176	1.0211	1.0247	1.0283	1.0319
46	1.0355	1.0391	1.0428	1.0464	1.0501	1.0538	1.0574	1.0611	1.0649	1.0686
47	1.0723	1.0761	1.0799	1.0837	1.0875	1.0913	1.0951	1.0990	1.1028	1.1067
48	1.1106	1.1145	1.1184	1.1223	1.1263	1.1303	1.1342	1.1382	1.1423	1.1463
49	1.1503	1.1544	1.1585	1.1626	1.1667	1.1708	1.1750	1.1791	1.1833	1.1875
50	1.1917	1.1960	1.2002	1.2045	1.2088	1.2131	1.2174	1.2217	1.2261	1.2305
51	1.2349	1.2393	1.2437	1.2482	1.2526	1.2571	1.2616	1.2662	1.2707	1.2753
52	1.2799	1.2845	1.2892	1.2938	1.2985	1.3032	1.3079	1.3126	1.3174	1.3222
53	1.3270	1.3318	1.3367	1.3416	1.3465	1.3514	1.3563	1.3613	1.3663	1.3713
54	1.3763	1.3814	1.3865	1.3916	1.3967	1.4019	1.4071	1.4123	1.4175	1.4228
55	1.4281	1.4334	1.4388	1.4441	1.4495	1.4550	1.4604	1.4659	1.4714	1.4769
56	1.4825	1.4881	1.4937	1.4994	1.5051	1.5108	1.5165	1.5223	1.5281	1.5339
57	1.5398	1.5457	1.5516	1.5576	1.5636	1.5696	1.5757	1.5818	1.5879	1.5941
58	1.6003	1.6065	1.6128	1.6191	1.6254	1.6318	1.6382	1.6446	1.6511	1.6577
59	1.6642	1.6708	1.6774	1.6841	1.6908	1.6976	1.7044	1.7112	1.7181	1.7250
60	1.7320	1.7390	1.7460	1.7531	1.7602	1.7674	1.7746	1.7819	1.7892	1.7966
61	1.8040	1.8114	1.8189	1.8265	1.8341	1.8417	1.8494	1.8571	1.8649	1.8728
62	1.8806	1.8886	1.8966	1.9046	1.9127	1.9209	1.9291	1.9374	1.9457	1.9541
63	1.9625	1.9710	1.9796	1.9882	1.9969	2.0056	2.0144	2.0233	2.0322	2.0412
64	2.0502	2.0593	2.0685	2.0777	2.0871	2.0964	2.1059	2.1154	2.1250	2.1347
65	2.1444	2.1542	2.1641	2.1740	2.1841	2.1942	2.2044	2.2146	2.2250	2.2354
66	2.2459	2.2565	2.2672	2.2779	2.2888	2.2997	2.3107	2.3218	2.3330	2.3443
67	2.3557	2.3672	2.3788	2.3904	2.4022	2.4141	2.4260	2.4381	2.4503	2.4626
68	2.4749	2.4874	2.5000	2.5127	2.5256	2.5385	2.5515	2.5647	2.5780	2.5914
69	2.6049	2.6186	2.6324	2.6463	2.6603	2.6745	2.6888	2.7032	2.7177	2.7325
70	2.7473	2.7623	2.7774	2.7927	2.8081	2.8237	2.8395	2.8554	2.8714	2.8876
71	2.9040	2.9206	2.9373	2.9542	2.9712	2.9885	3.0059	3.0235	3.0413	3.0593
72	3.0775	3.0958	3.1144	3.1332	3.1522	3.1714	3.1908	3.2104	3.2302	3.2503
73	3.2706	3.2911	3.3119	3.3329	3.3542	3.3757	3.3974	3.4195	3.4417	3.4643
74	3.4871	3.5102	3.5336	3.5573	3.5813	3.6056	3.6302	3.6551	3.6803	3.7058
75	3.7317	3.7579	3.7845	3.8114	3.8387	3.8664	3.8944	3.9228	3.9516	3.9808
76	4.0104	4.0404	4.0709	4.1018	4.1331	4.1649	4.1971	4.2299	4.2631	4.2968
77	4.3310	4.3658	4.4011	4.4369	4.4733	4.5102	4.5478	4.5859	4.6247	4.6641
78	4.7041	4.7448	4.7862	4.8283	4.8711	4.9146	4.9589	5.0039	5.0498	5.0964
79	5.1439	5.1923	5.2415	5.2917	5.3428	5.3948	5.4479	5.5019	5.5570	5.6132
80	5.6705	5.7289	5.7886	5.8494	5.9115	5.9749	6.0396	6.1057	6.1733	6.2423
81	6.3128	6.3849	6.4586	6.5340	6.6111	6.6901	6.7709	6.8536	6.9383	7.0252
82	7.1141	7.2053	7.2989	7.3948	7.4933	7.5943	7.6981	7.8047	7.9143	8.0269
83	8.1427	8.2619	8.3845	8.5108	8.6409	8.7750	8.9132	9.0559	9.2031	9.3551
84	9.5121	9.6745	9.8424	10.0162	10.1962	10.3827	10.5761	10.7768	10.9851	11.2016
85	11.4268	11.6611	11.9051	12.1595	12.4249	12.7021	12.9919	13.2951	13.6127	13.9458
86	14.2955	14.6631	15.0500	15.4578	15.8881	16.3431	16.8247	17.3355	17.8782	18.4558
87	19.0718	19.7303	20.4358	21.1934	22.0093	22.8903	23.8447	24.8819	26.0133	27.2523
88	28.6151	30.1212	31.7944	33.6642	35.7674	38.1507	40.8740	44.0157	47.6803	52.0102
θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

餘切$\cot\theta = \tan(90° - \theta)$

表 3. 360°與弧度換算表

θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	0.0000	0.0017	0.0035	0.0052	0.0070	0.0087	0.0105	0.0122	0.0140	0.0157
1	0.0175	0.0192	0.0209	0.0227	0.0244	0.0262	0.0279	0.0297	0.0314	0.0332
2	0.0349	0.0367	0.0384	0.0401	0.0419	0.0436	0.0454	0.0471	0.0489	0.0506
3	0.0524	0.0541	0.0558	0.0576	0.0593	0.0611	0.0628	0.0646	0.0663	0.0681
4	0.0698	0.0716	0.0733	0.0750	0.0768	0.0785	0.0803	0.0820	0.0838	0.0855
5	0.0873	0.0890	0.0908	0.0925	0.0942	0.0960	0.0977	0.0995	0.1012	0.1030
6	0.1047	0.1065	0.1082	0.1100	0.1117	0.1134	0.1152	0.1169	0.1187	0.1204
7	0.1222	0.1239	0.1257	0.1274	0.1292	0.1309	0.1326	0.1344	0.1361	0.1379
8	0.1396	0.1414	0.1431	0.1449	0.1466	0.1484	0.1501	0.1518	0.1536	0.1553
9	0.1571	0.1588	0.1606	0.1623	0.1641	0.1658	0.1675	0.1693	0.1710	0.1728
10	0.1745	0.1763	0.1780	0.1798	0.1815	0.1833	0.1850	0.1867	0.1885	0.1902
11	0.1920	0.1937	0.1955	0.1972	0.1990	0.2007	0.2025	0.2042	0.2059	0.2077
12	0.2094	0.2112	0.2129	0.2147	0.2164	0.2182	0.2199	0.2217	0.2234	0.2251
13	0.2269	0.2286	0.2304	0.2321	0.2339	0.2356	0.2374	0.2391	0.2409	0.2426
14	0.2443	0.2461	0.2478	0.2496	0.2513	0.2531	0.2548	0.2566	0.2583	0.2600
15	0.2618	0.2635	0.2653	0.2670	0.2688	0.2705	0.2723	0.2740	0.2758	0.2775
16	0.2792	0.2810	0.2827	0.2845	0.2862	0.2880	0.2897	0.2915	0.2932	0.2950
17	0.2967	0.2984	0.3002	0.3019	0.3037	0.3054	0.3072	0.3089	0.3107	0.3124
18	0.3142	0.3159	0.3176	0.3194	0.3211	0.3229	0.3246	0.3264	0.3281	0.3299
19	0.3316	0.3334	0.3351	0.3368	0.3386	0.3403	0.3421	0.3438	0.3456	0.3473
20	0.3491	0.3508	0.3526	0.3543	0.3560	0.3578	0.3595	0.3613	0.3630	0.3648
21	0.3665	0.3683	0.3700	0.3717	0.3735	0.3752	0.3770	0.3787	0.3805	0.3822
22	0.3840	0.3857	0.3875	0.3892	0.3909	0.3927	0.3944	0.3962	0.3979	0.3997
23	0.4014	0.4032	0.4049	0.4067	0.4084	0.4101	0.4119	0.4136	0.4154	0.4171
24	0.4189	0.4206	0.4224	0.4241	0.4259	0.4276	0.4293	0.4311	0.4328	0.4346
25	0.4363	0.4381	0.4398	0.4416	0.4433	0.4451	0.4468	0.4485	0.4503	0.4520
26	0.4538	0.4555	0.4573	0.4590	0.4608	0.4625	0.4642	0.4660	0.4677	0.4695
27	0.4712	0.4730	0.4747	0.4765	0.4782	0.4800	0.4817	0.4834	0.4852	0.4869
28	0.4887	0.4904	0.4922	0.4939	0.4957	0.4974	0.4992	0.5009	0.5026	0.5044
29	0.5061	0.5079	0.5096	0.5114	0.5131	0.5149	0.5166	0.5184	0.5201	0.5218
30	0.5236	0.5253	0.5271	0.5288	0.5306	0.5323	0.5341	0.5358	0.5376	0.5393
31	0.5410	0.5428	0.5445	0.5463	0.5480	0.5498	0.5515	0.5533	0.5550	0.5568
32	0.5585	0.5602	0.5620	0.5637	0.5655	0.5672	0.5690	0.5707	0.5725	0.5742
33	0.5759	0.5777	0.5794	0.5812	0.5829	0.5847	0.5864	0.5882	0.5899	0.5917
34	0.5934	0.5951	0.5969	0.5986	0.6004	0.6021	0.6039	0.6056	0.6074	0.6091
35	0.6109	0.6126	0.6143	0.6161	0.6178	0.6196	0.6213	0.6231	0.6248	0.6266
36	0.6283	0.6301	0.6318	0.6335	0.6353	0.6370	0.6388	0.6405	0.6423	0.6440
37	0.6458	0.6475	0.6493	0.6510	0.6527	0.6545	0.6562	0.6580	0.6597	0.6615
38	0.6632	0.6650	0.6667	0.6684	0.6702	0.6719	0.6737	0.6754	0.6772	0.6789
39	0.6807	0.6824	0.6842	0.6859	0.6876	0.6894	0.6911	0.6929	0.6946	0.6964
40	0.6981	0.6999	0.7016	0.7034	0.7051	0.7068	0.7086	0.7103	0.7121	0.7138
41	0.7156	0.7173	0.7191	0.7208	0.7226	0.7243	0.7260	0.7278	0.7295	0.7313
42	0.7330	0.7348	0.7365	0.7383	0.7400	0.7418	0.7435	0.7452	0.7470	0.7487
43	0.7505	0.7522	0.7540	0.7557	0.7575	0.7592	0.7610	0.7627	0.7644	0.7662
44	0.7679	0.7697	0.7714	0.7732	0.7749	0.7767	0.7784	0.7801	0.7819	0.7836
45	0.7854	0.7871	0.7889	0.7906	0.7924	0.7941	0.7959	0.7976	0.7993	0.8011
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9

表 3. （續）360°與弧度換算表

θ (degrees)	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
45	0.7854	0.7871	0.7889	0.7906	0.7924	0.7941	0.7959	0.7976	0.7993	0.8011
46	0.8028	0.8046	0.8063	0.8081	0.8098	0.8116	0.8133	0.8151	0.8168	0.8185
47	0.8203	0.8220	0.8238	0.8255	0.8273	0.8290	0.8308	0.8325	0.8343	0.8360
48	0.8377	0.8395	0.8412	0.8430	0.8447	0.8465	0.8482	0.8500	0.8517	0.8535
49	0.8552	0.8569	0.8587	0.8604	0.8622	0.8639	0.8657	0.8674	0.8692	0.8709
50	0.8726	0.8744	0.8761	0.8779	0.8796	0.8814	0.8831	0.8849	0.8866	0.8884
51	0.8901	0.8918	0.8936	0.8953	0.8971	0.8988	0.9006	0.9023	0.9041	0.9058
52	0.9076	0.9093	0.9110	0.9128	0.9145	0.9163	0.9180	0.9198	0.9215	0.9233
53	0.9250	0.9268	0.9285	0.9302	0.9320	0.9337	0.9355	0.9372	0.9390	0.9407
54	0.9425	0.9442	0.9460	0.9477	0.9494	0.9512	0.9529	0.9547	0.9564	0.9582
55	0.9599	0.9617	0.9634	0.9652	0.9669	0.9686	0.9704	0.9721	0.9739	0.9756
56	0.9774	0.9791	0.9809	0.9826	0.9843	0.9861	0.9878	0.9896	0.9913	0.9931
57	0.9948	0.9966	0.9983	1.0001	1.0018	1.0035	1.0053	1.0070	1.0088	1.0105
58	1.0123	1.0140	1.0158	1.0175	1.0193	1.0210	1.0227	1.0245	1.0262	1.0280
59	1.0297	1.0315	1.0332	1.0350	1.0367	1.0385	1.0402	1.0419	1.0437	1.0454
60	1.0472	1.0489	1.0507	1.0524	1.0542	1.0559	1.0577	1.0594	1.0611	1.0629
61	1.0646	1.0664	1.0681	1.0699	1.0716	1.0734	1.0751	1.0769	1.0786	1.0803
62	1.0821	1.0838	1.0856	1.0873	1.0891	1.0908	1.0926	1.0943	1.0960	1.0978
63	1.0995	1.1013	1.1030	1.1048	1.1065	1.1083	1.1100	1.1118	1.1135	1.1152
64	1.1170	1.1187	1.1205	1.1222	1.1240	1.1257	1.1275	1.1292	1.1310	1.1327
65	1.1344	1.1362	1.1379	1.1397	1.1414	1.1432	1.1449	1.1467	1.1484	1.1502
66	1.1519	1.1536	1.1554	1.1571	1.1589	1.1606	1.1624	1.1641	1.1659	1.1676
67	1.1694	1.1711	1.1728	1.1746	1.1763	1.1781	1.1798	1.1816	1.1833	1.1851
68	1.1868	1.1885	1.1903	1.1920	1.1938	1.1955	1.1973	1.1990	1.2008	1.2025
69	1.2043	1.2060	1.2077	1.2095	1.2112	1.2130	1.2147	1.2165	1.2182	1.2200
70	1.2217	1.2235	1.2252	1.2269	1.2287	1.2304	1.2322	1.2339	1.2357	1.2374
71	1.2392	1.2409	1.2427	1.2444	1.2461	1.2479	1.2496	1.2514	1.2531	1.2549
72	1.2566	1.2584	1.2601	1.2619	1.2636	1.2653	1.2671	1.2688	1.2706	1.2723
73	1.2741	1.2758	1.2776	1.2793	1.2811	1.2828	1.2845	1.2863	1.2880	1.2898
74	1.2915	1.2933	1.2950	1.2968	1.2985	1.3002	1.3020	1.3037	1.3055	1.3072
75	1.3090	1.3107	1.3125	1.3142	1.3160	1.3177	1.3194	1.3212	1.3229	1.3247
76	1.3264	1.3282	1.3299	1.3317	1.3334	1.3352	1.3369	1.3386	1.3404	1.3421
77	1.3439	1.3456	1.3474	1.3491	1.3509	1.3526	1.3544	1.3561	1.3578	1.3596
78	1.3613	1.3631	1.3648	1.3666	1.3683	1.3701	1.3718	1.3736	1.3753	1.3770
79	1.3788	1.3805	1.3823	1.3840	1.3858	1.3875	1.3893	1.3910	1.3927	1.3945
80	1.3962	1.3980	1.3997	1.4015	1.4032	1.4050	1.4067	1.4085	1.4102	1.4119
81	1.4137	1.4154	1.4172	1.4189	1.4207	1.4224	1.4242	1.4259	1.4277	1.4294
82	1.4311	1.4329	1.4346	1.4364	1.4381	1.4399	1.4416	1.4434	1.4451	1.4469
83	1.4486	1.4503	1.4521	1.4538	1.4556	1.4573	1.4591	1.4608	1.4626	1.4643
84	1.4661	1.4678	1.4695	1.4713	1.4730	1.4748	1.4765	1.4783	1.4800	1.4818
85	1.4835	1.4853	1.4870	1.4887	1.4905	1.4922	1.4940	1.4957	1.4975	1.4992
86	1.5010	1.5027	1.5044	1.5062	1.5079	1.5097	1.5114	1.5132	1.5149	1.5167
87	1.5184	1.5202	1.5219	1.5236	1.5254	1.5271	1.5289	1.5306	1.5324	1.5341
88	1.5359	1.5376	1.5394	1.5411	1.5428	1.5446	1.5463	1.5481	1.5498	1.5516
89	1.5533	1.5551	1.5568	1.5586	1.5603	1.5620	1.5638	1.5655	1.5673	1.5690
90	1.5708	1.5725	1.5743	1.5760	1.5778	1.5795	1.5812	1.5830	1.5847	1.5865
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9